

Do Robots Really Need Anthropomorphic Hands?

A Summary of a Comparison of Human and Robotic Hands

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Abstract

The human hand is widely used as the benchmark for dexterous robotic end-effector design, yet the necessity of anthropomorphic morphology is challenged by competition outcomes and tool-use studies. We combine a comparative analysis of human and robotic hand capabilities with a systematic literature review of 125 manipulation studies (2019–2025) [1] to evaluate this assumption. We find that mechanism complexity predicts the breadth of demonstrated skill repertoires but not the depth of in-hand manipulation, motivating a shift from form-based toward function-based biomimicry. **This abstract summarizes work currently under review and available as an arXiv preprint [1].**

1. Introduction

The human hand is often treated as the gold standard for robotic manipulation. Yet competition outcomes tell a different story. The Amazon Picking Challenge winner used suction, no DARPA Robotics Challenge team deployed a fully actuated anthropomorphic hand, and the Cybathlon Powered Arm Race winner used a body-powered hook [2]. Conversely, individuals with six functional fingers solve tasks one-handed that typical five-fingered humans need both hands for [3], and birds manipulate complex objects with their bills alone [4]. We address the question through a comparative analysis of human and robotic hand capabilities and a systematic literature review correlating hand features with demonstrated manipulation skills.

2. The Human Hand

Biomechanics. The human hand has five fingers and approximately 24 DoF [5]. Yet most grasps engage only three fingers for load bearing [6], and precision grasps rely more on finger abduction/adduction than on additional flexion/extension DoF [7]. Wrist flexibility matters more than finger count: restricting the wrist to pronation/supination impairs SHAP benchmark performance as much as reducing the hand to open/close, a loss of 21 finger DoF [8]. The soft, deformable palm affords passive shape adaptation and enables exploitation of environmental contacts during manipulation [9].

Perception. The hand integrates tactile, proprioceptive, and visual signals across cutaneous, muscular, and articular receptors [10, 11]. Four mechanoreceptor types provide spatial acuity down to 0.5 mm and cover frequencies from slow texture detection to high-frequency vibration. Proprioceptive afferents conduct at 80–120 m/s and are essential for coordinated movement: proprioceptive impairment alone degrades manipulation trajectories even when visual feedback is available [11].

3. Robotic Hands

Mechanics. Robotic hand designs span six kinematic architectures from fully actuated to underactuated and redundant [12]. Configurations range from two-finger parallel grippers and three-fingered hands (e.g., BarrettHand) to four-fingered non-anthropomorphic hands with two opposing finger pairs [13] and five-fingered designs such as the 24-DoF Shadow Dexterous Hand. Semi-rigid hands with compliant flexure joints [14] provide passive shape adaptation between fully rigid and fully soft architectures. Current trends include AI-driven hardware/learning

co-design, quasi-direct-drive actuation [15, 16], and compliant materials [17]. The most advanced hands match the human hand in DoF and actuator counts but lack its softness, deformability, and passive range of motion, and dexterity is typically traded against force production.

Perception. Tactile sensors mimic human mechanoreceptors using a variety of transduction technologies [18]. Commercial sensors such as GelSight/DIGIT [19] and Xela uSkin [20] provide sub-millimetre spatial resolution and multi-axis force readout. Yet robotic hands rely on relatively few, often bulky sensors whose data require considerable post-processing before they can drive control, while thermoception and nociception are barely represented in hardware.

4. Systematic Review

We searched Google Scholar and IEEE Xplore (2019–2025), selecting 125 papers from 4,325 records that demonstrate multi-finger skills beyond simple pinch grasps. We extracted mechanism features, sensor setup, and demonstrated skill categories from Dollar’s 15-category manipulation taxonomy [21]. We test two hypotheses: (H1) skill repertoire size and in-hand manipulation depth increase with mechanism complexity (number of fingers, DoF, actuators, presence of abduction/adduction), and (H2) skill repertoire size increases with tactile sensing and the number of integrated sensor modalities. We additionally compare the mechanical complexity of surveyed hands to the human hand and quantify the breadth of skill categories demonstrated per hand. Tests use Spearman ρ for count features and Mann-Whitney U with rank-biserial r_{rb} for binary features, jointly corrected with Benjamini–Hochberg FDR ($\alpha = 0.05$) across all 15 tests.

5. Discussion and Conclusion

Mechanism complexity matches the human hand. Two five-fingered designs reach DoF and actuator counts comparable to the human hand: the Shadow Dexterous Hand and the bio-inspired hand of Zhou et al. [22]. Both implement independent abduction/adduction of fingers 2–5, and Zhou et al.’s hand additionally reproduces palm folding [1]. What current hands still lack are the softness, deformability, and passive range of motion of human joints, properties tied to robust contact-rich manipulation rather than to morphology.

Hardware predicts breadth, not depth. After aggregating results per hand ($N = 93$), mechanism complexity correlates positively with skill repertoire breadth: fingers $\rho_s = .44$, DoF $\rho_s = .50$, actuators $\rho_s = .37$ (all $p^* \leq .001$, medium-strength effects); hands with abduction/adduction capability ($r_{rb} = .48$, $p^* < .001$) and kinesthetic sensing ($r_{rb} = .44$, $p^* < .001$) demonstrate significantly broader skill repertoires [1]. By contrast, no feature predicts the DoF controlled during in-hand manipulation (all non-significant after FDR correction).

Simple mechanisms suffice for in-hand manipulation. A two-finger, single-actuator gripper achieves 3-DoF in-hand manipulation by exploiting environmental contacts as a virtual third finger [23]. Three fingers with four actuators [24] or four fingers in two opposing pairs [25] enable 6-DoF control. Manipulation depth is gated by control strategy, not hardware morphology.

Table 1: Recommendations (top) and open research questions (bottom).

Feature	Finding
Finger count	2–3 sufficient for most grasps; 3–4 fingers with 4 actuators can manipulate 6 object DoF [24, 25]
Finger joints	Abduction/adduction extends skill repertoire ($r_{rb} = .48$)
Evaluation	Standardised cross-category benchmarks needed
Feature	Open question
Arm	Is a redundant 7-DoF arm more beneficial than a complex hand?
Surface	Does compliant cover improve robustness and enable learning?
Control	To what extent can intelligent control compensate for simpler hardware?

Sensor integration lags sensor availability. While 22 surveyed manipulation systems integrate all four evaluated sensor modalities, only 18 of 62 studies using proprioception add tactile sensing, and 7 of those treat it as a binary contact signal rather than exploiting its full resolution [1]. The field is also bifurcated between minimal grippers and anthropomorphic designs, with intermediate non-anthropomorphic morphologies such as three-fingered hands or two opposing finger pairs [13] remaining underexplored.

Function-based biomimicry. The human hand’s key advantage lies not in its form but in functional properties that robotic hands have not yet replicated: mechanical compliance enabling contact-rich exploration and learning, wrist mobility providing a larger performance gain than finger DoF count [8], and adaptive sensorimotor control. Anthropomorphic morphology may even be counterproductive in HRI due to the uncanny valley [26].

Beyond the human hand. No surveyed paper uses more than 22 DoF or actuators, and only one uses more than five fingers [1]. Yet three fingers with four actuators [24] or four fingers in two opposing pairs [25] already control all six DoF of an object, and environmental contacts can substitute for additional digits [23]. Non-anthropomorphic designs with multiple opposing finger pairs or with more than five digits may therefore outperform the human hand on parallel manipulation and multi-object grasping.

Recommendations and future work. Table 1 consolidates our findings as design recommendations and open questions. Two limitations of our review motivate a call for better benchmarks: skill demonstrations are recorded as binary with no measure of proficiency, and complex hands are tested in only a narrow subset of Dollar’s taxonomy [21], almost certainly underreporting their true capability and limiting our ability to determine whether anthropomorphic designs confer any demonstrable advantage. Standardised cross-category benchmarks quantifying proficiency, not merely task completion, are therefore a precondition for robust evaluation. We release our dataset of hand–skill mappings to support benchmark development.¹

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¹<https://figshare.com/s/f4fdd9acalc134f010dd>